

Q2.



Renewal Cycle:

→ Begins when a job arrives to an empty system, and ends when a new job arrives at an empty system.

$$T = I + B$$

$\uparrow$ $\uparrow$ $\uparrow$
 total len of cycle idle len Busy period.

$$E(I) = \frac{1}{\lambda_1 + \lambda_2} \quad \leftarrow (1)$$

$$EB = E(B | \text{Stream 1}) \cdot P(\text{job recievng service was of type 1}) + E(B | \text{Stream 2}) \cdot P(\text{Stream 2})$$

→ Let μ_1, μ_2 be average service time of server for Stream 1 & 2.

→ Let p_1, p_2 be probabilities that a job receiving service is of type 1 & 2 resp (Equal to probs that a randomly arrived job to an empty system is of type 1 & 2)

$$p_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

$$p_2 = \frac{\lambda_2}{\lambda_1 + \lambda_2}$$

Proof: Streams 1 and 2 arrive with rates λ_1, λ_2 .
 Let x_1, x_2 be times until next arrivals.

$$\begin{aligned}
 P(x_1 < x_2) &= \int_0^{\infty} P(x_1 < x_2 | x_2 = x_2) f_{x_2}(x_2) dx_2 \\
 &= \int_0^{\infty} (1 - e^{-\lambda_1 x}) \lambda_2 e^{-\lambda_2 x} dx \\
 &= \int_0^{\infty} \lambda_2 e^{-\lambda_2 x} dx - \lambda_1 \int_0^{\infty} e^{-(\lambda_1 + \lambda_2)x} dx \\
 &= 1 - \frac{\lambda_2}{\lambda_1 + \lambda_2} = \frac{\lambda_1}{\lambda_1 + \lambda_2}
 \end{aligned}$$

$$EB = \frac{\lambda_1 \mu_1 + \lambda_2 \mu_2}{\lambda_1 + \lambda_2} \quad \leftarrow (2)$$

$$\Rightarrow ET = \frac{\lambda_1 \mu_1 + \lambda_2 \mu_2 + 1}{\lambda_1 + \lambda_2} \quad \leftarrow (3)$$

Defining Rewards.

Goal: Find Busy periods B_1 & B_2 for streams 1 & 2.

$\therefore R_i \Rightarrow$ reward associated with any arrival cycle is time spent by server in servicing a job of type i .

Any given cycle could have had either of the jobs.

So; $ER_i = \left(\text{prob. that cycle had job of type } i \right) \times \mu_i$

$$\Rightarrow ER_1 = p_1 \mu_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2} \mu_1$$

$$\Rightarrow ER_2 = p_2 \mu_2 = \frac{\lambda_2}{\lambda_1 + \lambda_2} \mu_2$$

From renewal reward theorem for Goal 1 (Reward is R_1):

$$\lim_{t \rightarrow \infty} \frac{R_1(t)}{t} = \frac{ER_1}{ET}$$



Long-run percent of time spent servicing jobs of type 1.
(Time average occupancy).

$$B_1 = \frac{\frac{\lambda_1 \mu_1}{\lambda_1 + \lambda_2}}{\frac{\lambda_1 \mu_1 + \lambda_2 \mu_2 + 1}{\lambda_1 + \lambda_2}}$$

$$B_1 = \frac{\lambda_1 \mu_1}{\lambda_1 \mu_1 + \lambda_2 \mu_2 + 1}$$

Similarly;

$$B_2 = \frac{\lambda_2 \mu_2}{\lambda_1 \mu_1 + \lambda_2 \mu_2 + 1}$$

Note: By ergodicity, $\left(\frac{\lambda_1 \mu_1}{\lambda_1 \mu_1 + \lambda_2 \mu_2 + 1}, \frac{\lambda_2 \mu_2}{\lambda_1 \mu_1 + \lambda_2 \mu_2 + 1}, \frac{1}{\lambda_1 \mu_1 + \lambda_2 \mu_2 + 1} \right)$

are also stationary probabilities of being in the three possible states: occupied with job 1, 2 and being empty at steady state.

Q3. M/G/1.

Say the service time follows distribution $F(\cdot)$



From Lec 11; we have,

$$W = w + b$$

$$L = l + \rho$$

$$w = \frac{\lambda M_2}{2(1-\rho)}$$

(refer to slides for meaning).

By Little's Law;

$$l = \lambda w = \frac{\lambda^2 M_2}{2(1-\rho)}$$

$$L = \frac{\lambda^2 M_2}{2(1-\rho)} + \rho$$

$$1) X(t) = A(t) - D(t)$$

where $A(t)$, $D(t)$ are poisson processes with rates λ & μ respectively.

→ A renewal process is a counting process for which the inter arrival times are i.i.d with an arbitrary distribution.

→ $\{X(t), t \geq 0\}$ is not a counting process cuz

$$t < s \nRightarrow X(t) < X(s)$$

→ $X(t)$ is a CTMC. Consider $t_1 < t_2 < \dots < t_n$

$$P(X(t_n) = x_n | X(t_1) = x_1, X(t_2) = x_2, \dots, X(t_{n-1}) = x_{n-1})$$

$$= P(A(t_n) - D(t_n) = x_n | A(t_1) - D(t_1) = x_1, \dots, A(t_{n-1}) - D(t_{n-1}) = x_{n-1})$$

Since $\{A(t)\}$ & $\{D(t)\}$ are ^{independent poisson} counting processes, they satisfy the markov property. So does $A(t) - D(t)$

$$= P(A(t_n) - D(t_n) = x_n | A(t_{n-1}) - D(t_{n-1}) = x_{n-1})$$

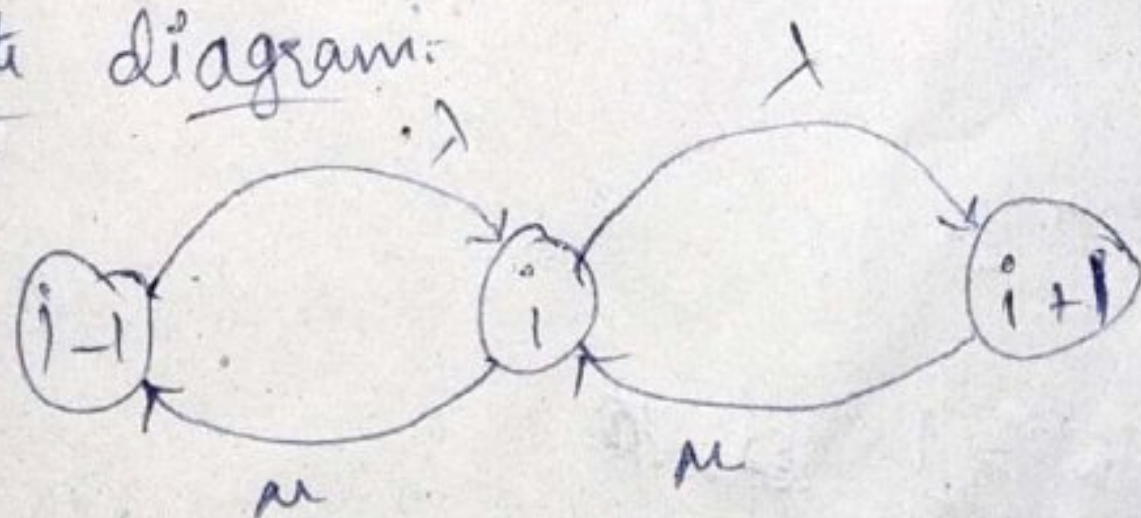
$$= P(X(t_n) = x_n | X(t_{n-1}) = x_{n-1})$$

∴ $X(t)$ is a markov chain and the support set

$t \in \mathbb{R}$ is continuous

∴ $X(t)$ is a CTMC

State diagram:



$$\pi_i(\lambda + \mu) = \pi_{i+1}(\mu) + \pi_{i-1}(\lambda)$$

$$\Rightarrow \pi_i \lambda + \pi_i \mu = \pi_{i+1} \mu + \pi_{i-1} \lambda$$

$$\Rightarrow \pi_{i+1} = \rho \pi_i \text{ and } \pi_i = \rho \pi_{i-1}$$

$$[Q]_{ij} = \begin{cases} -(\lambda + \mu) & i=j \\ \lambda & j=i+1 \\ \mu & j=i-1 \\ 0 & \text{otherwise} \end{cases}$$

$Q \rightarrow$ infinite dimensional matrix.

Stationary distribution

Case 1: $\lambda > \mu \Rightarrow \rho > 1$

$$\pi_{i+1} = \rho \pi_i$$

$$\pi_n = \rho^n \pi_0$$

$$\lim_{n \rightarrow \infty} \pi_n \rightarrow \infty$$

$\therefore$ Stationary distribution doesn't exist

Case 2: $\lambda < \mu \Rightarrow \rho < 1$

$$\pi_n = \rho^n \pi_0$$

$$\lim_{n \rightarrow \infty} \pi_n \rightarrow 0$$

$\therefore$ Stationary distribution doesn't exist

Case 3: $\lambda = \mu$

$$\pi_n = \pi_0 \quad \forall n$$

$$\sum_{n=-\infty}^{\infty} \pi_n = 1$$

$$\text{if } \pi_n > 0, \text{ LHS} \rightarrow \infty$$

$$\pi_n = 0, \text{ LHS} = 0$$

$\therefore$ no ~~such~~ stationary distribution exists which satisfies $\sum_{n=-\infty}^{\infty} \pi_n = 1$

$N(t) = \max(X(t), 0)$ is not a Markov chain because

$$P(N(t+h)=1 \mid N(t)=0, \mathcal{F}_t^N) \neq P(N(t+h)=1 \mid N(t)=0)$$

where $\mathcal{F}_t^N$ encodes the ^{entire} information ^{of $N(t)$} from ~~$t=0$ to t~~ $t \in [0, t)$

→ ^{Given t} Depending on $N(t) > 0$ ~~$N(t)$~~

$X(t)$ can be ~~either~~ any value ≤ 0

cannot be determined by $N(t)$ alone.

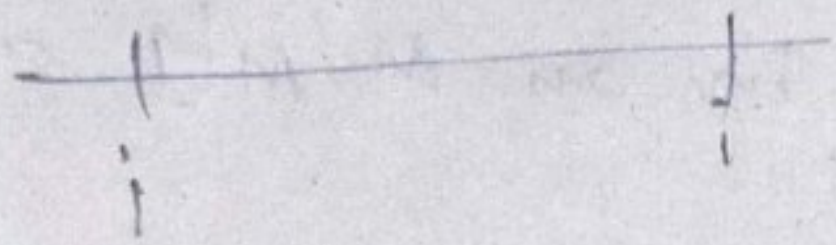
→ $N(t) > 0 \Rightarrow X(t) \leq 0$ but $N(t+h)$ requires $X(t)$ (hidden information) which might be determined using few kinds of $\mathcal{F}_t^N$.

⇒ future doesn't depend only on the present

Considering the descriptor is $N(t)$

→ If descriptor is $\{(X(t), N(t)), t \geq 0\}$, then this is a Markov chain.

4) Let $X(t)$ denote the CTMC



Renewal: The markov chain visiting state i ^{consecutive}
 X_i : Cycle length Time between 2nd visits to the markov chain to the state i

$$E[X_i] \neq \infty$$

define $X_{ij} \rightarrow$ Time between $j-1^{th}$ and j^{th} arrival
 $\downarrow$
 Length of the renewal cycle of the markov chain to state i

$$\Rightarrow E[X_{ij}] = \mu_{ij} \quad \forall j \in N$$

$$\text{Reward: } r(t) = \begin{cases} 1 & \text{if } X(t) = i \\ 0 & \text{else} \end{cases}$$

$\rightarrow$ Reward in one cycle is the time for which the markov chain is in state i . Since the renewal cycle ends after the next visit to state i , reward accumulated in that cycle is just the time taken by the markov chain to leave the state i .

$$\therefore E[R] = \frac{1}{\mu_{ii}}$$

$$R \sim \min(\exp(q_{ii}), \infty)$$

$$R \sim \exp(-q_{ii})$$

$$\therefore E[R] = \frac{1}{q_{ii}}$$

Using renewal reward theorem,

$$R(t) = \int_0^t r(t) dt$$

$$\lim_{t \rightarrow \infty} \frac{R(t)}{t} \xrightarrow{\text{a.s.}} \frac{E[R]}{E[X]} = \frac{1}{|Q_{ii}| \mu_{ii}}$$

but $\pi_i = \lim_{t \rightarrow \infty} \frac{R(t)}{t} \rightarrow$ average time spent in state i .

$$\pi_i = \frac{1}{|Q_{ii}| \mu_{ii}}$$